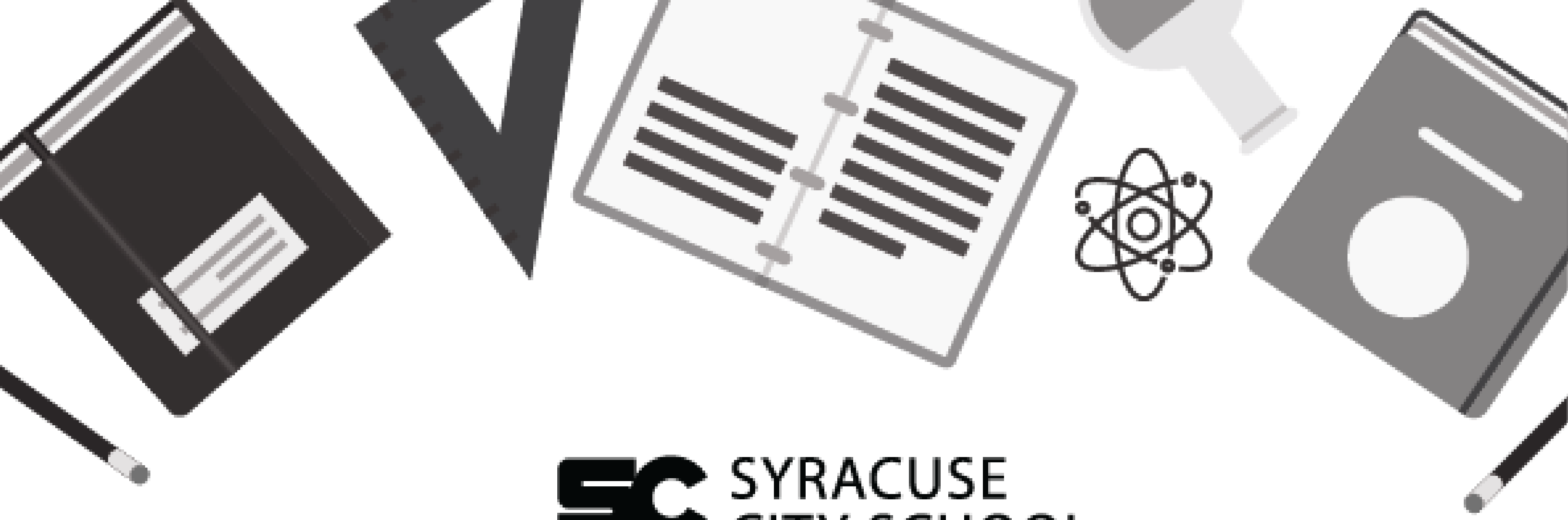

SYRACUSE CITY SCHOOL DISTRICT

DISTANCE LEARNING

Algebra 2

Coursework for May 2020

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The Functions of Algebra 2

Definition of Function: A function is a relationship where each input has a single output.

There are several types of functions, linear, exponential, quadratic, and polynomial, and their information is listed below.

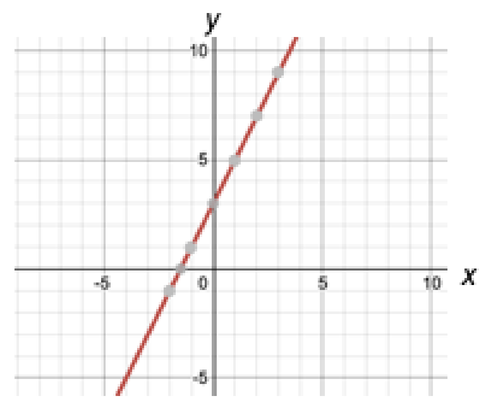
1. Linear Functions

Definition: Linear functions are any functions that create a straight line when graphed.

Representations:

Words	Formula/Symbolic	Table	Graph
Multiply x by 2 and add 3 to get y.	y = 2x + 3 or *f(x)* = 2x + 3	see table below	see graph below

x	Y
-2	-1
-1	1
0	3
1	5
2	7
3	9



Linear Function Formats:

There are a couple of formats for linear functions:

Slope-intercept form

$$y = \overbrace{m}^{\text{slope}}x + \underbrace{b}_{\text{y-intercept}}$$

Point-Slope form

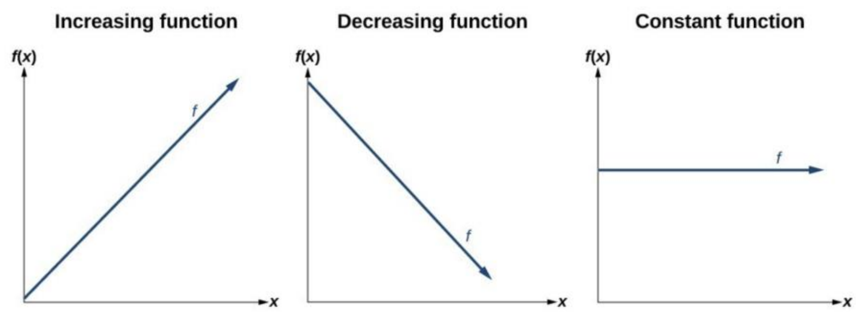
$$y - y_1 = \overbrace{m}^{\text{slope}}(x - x_1)$$

coordinates of a point on the line

Linear Function Tidbits:

Slope: The slope is the change y values divided by the change in x values between two points, and for linear functions, the slope (or rate of change) is constant.

Increasing and decreasing: the function is increasing if the line is rising as it moves to the right; it is decreasing if the line is falling as it moves to the right, and it is constant if the line is horizontal.



x-intercept: The point where the line crosses the x axis

y-intercept: The point where the line crosses the y axis

Test if it's a function: To be a function each input (x) can only have one output (y). If you look at the graph, and an x value has more than one point above or below it, then it is not a function. This is called the **Vertical Line Test** – if you can draw a vertical line and hit more than one point the graph is not a function.

2. Exponential Functions

Definition: Exponential functions are any functions that have a constant whose exponent is a variable. The constant must be greater than 0 (> 0) and cannot be 1.

Representations:

Words	**Formula/Symbolic**
To get y, we take the number 3 and raise it to the power x.	$f(x) = 3^x$ or $y = 3^x$

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The Functions of Algebra 2

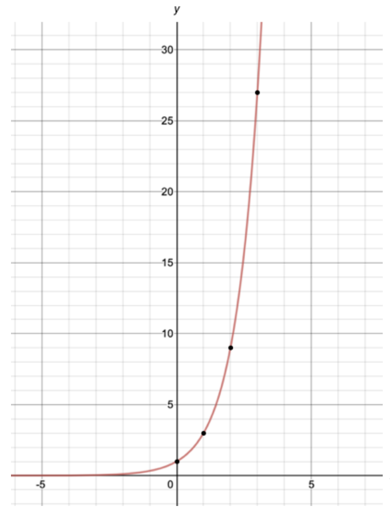
Exponential Functions Continued

Table

$y = 3^x$

x	Y
0	1
1	3
2	9
3	27
4	81

Graph



Exponential Function Format:

$$f(x) = a^x$$

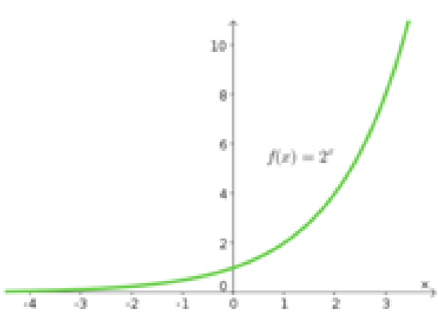
Exponent (often time)

Base (often starting amount)

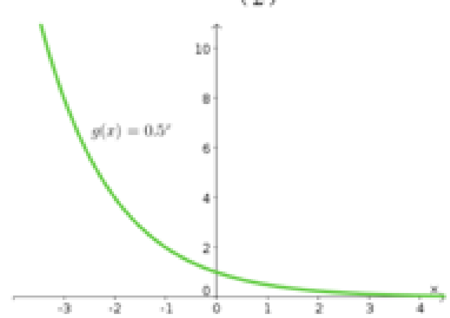
Exponential Function Tidbits:

Growth and Decay: When the base of an exponential function is greater than 1, then the function shows growth and is increasing. When the base is between 0 and 1, the function shows decay and is decreasing.

$f(x) = 2^x$.



$g(x) = \left(\frac{1}{2}\right)^x$



3. Polynomial Functions

Definition: Polynomial Functions are functions with one or more terms where the variables have non-negative exponents (0, 1, 2, etc). In other words, a polynomial is the sum of one or more terms with real coefficients and nonnegative integer exponents. The degree of the polynomial function is largest exponent present in the function.

Examples and non-examples:

These **are** polynomials:

- **$3x$** **(first degree polynomial)**
- **$x - 2$** **(first degree polynomial)**
- **$-6y^2 - (79)x$** **(second degree polynomial)**
- **$3xyz + 3xy^2z - 0.1xz - 200y + 0.5$** **(second degree polynomial)**
- **$512v^5 + 99w^5$** **(fifth degree polynomial)**
- **5**

(Yes, "5" is a polynomial, **one term is allowed**, and it can be just a constant!)

These are **not** polynomials

- **$3xy^{-2}$** is not, because the exponent is "-2" (exponents can only be 0,1,2,...)
- **$2/(x+2)$** is not, because dividing by a variable is not allowed
- **$1/x$** is not either
- $\sqrt{x}$ is not, because the exponent is "½"

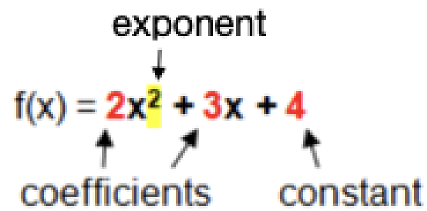
There are certain polynomials that we are more interested in. Specifically, 2nd degree polynomials called quadratics and 3rd degree polynomials called cubics.

3a. Quadratics

Definition: Quadratic functions are any functions in the form $\mathbf{f(x) = ax^2 + bx + c}$, where **a**, **b**, and **c** are numbers with **a** not equal to zero (b and c can be zero). Quadratics are a specific form of polynomials.

Representations:

Formula: To be a quadratic function there must be a variable with an exponent of 2 (a squared variable).

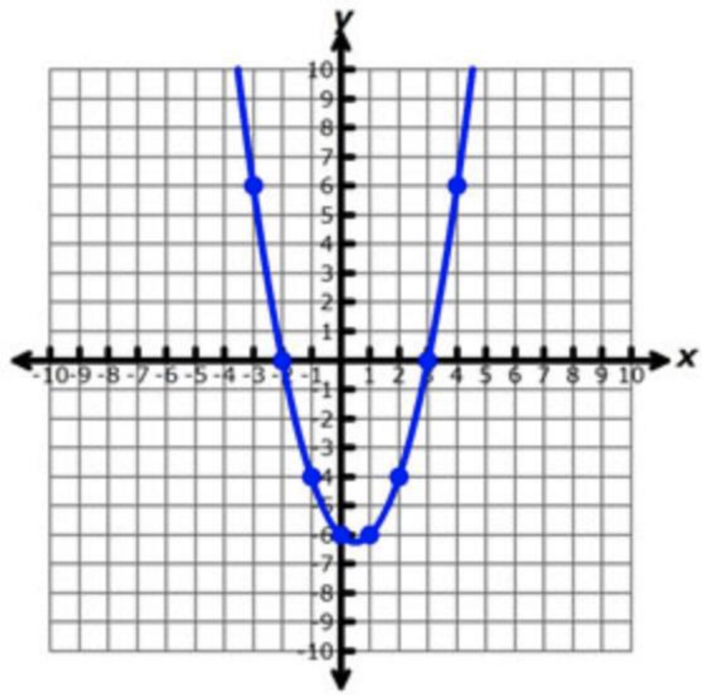


Table

$y = x^2 - x - 6$

x	y
−3	6
−2	0
−1	−4
0	−6
1	−6
2	−4
3	0
4	6

Graph



Quadratic Function Formats: Quadratic functions have two useful formats: general and vertex

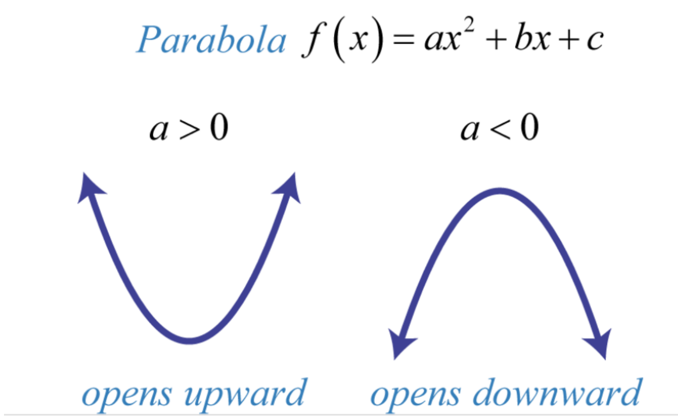
general form

$$f(x) = ax^2 + bx + c$$

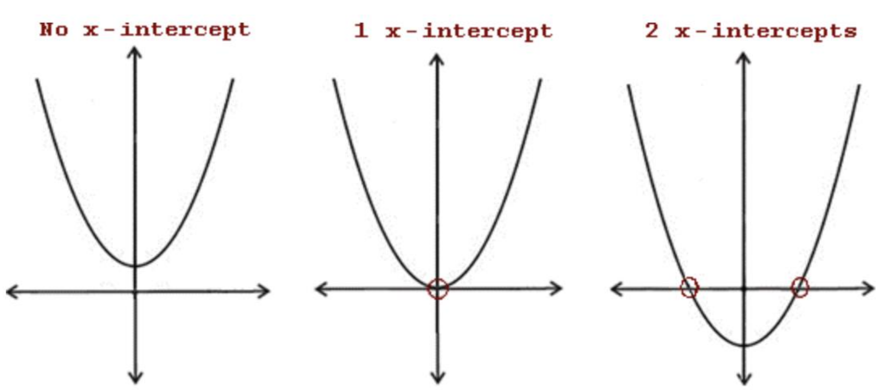
vertex form

$$f(x) = a(x-h)^2 + k$$

Quadratic Graphs: The graphs of quadratics create a parabola. Parabolas open upward (like a U) if the coefficient of the squared term (a) is positive. It turns downward if a is negative.



Intercepts - A parabola can intersect with the x axis in 1, 2, or no places. The place where the parabola meets the x axis is called a zero, because that is where the value of the function is zero.



Vertex - A parabola always has a vertex, and it is that point that is either highest or lowest on the parabola. If it is high, it is called a maximum, and if it is low it is called a minimum.

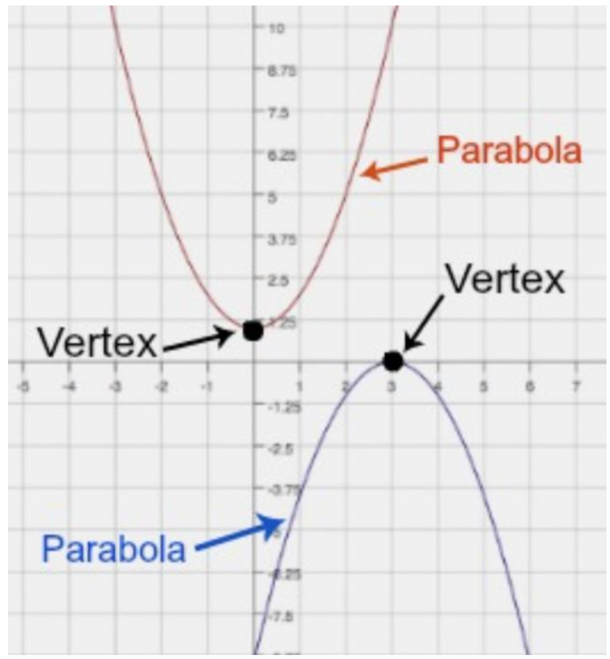
The points a vertex can be found by using the formula -b/2a to find the x value when the function is in standard (general) form. Substitute the number in for x and find the y value, giving you the ordered pair that is the vertex. If the formula is in vertex form, then h, k are the ordered pair for the vertex.

For example: $y = x^2 - 2x + 3$

$$\frac{-b}{2a} = \frac{2}{2}(1) = 1; \quad y = 1^2 - 2(1) + 3$$

$$y = 2$$

The vertex is (1, 2)



Finding Zeros - To find the zeros (the values of x that make the function =0), also called roots, you can graph the function and see where the parabola hits the x axis.

You can also factor the quadratic to find the roots.

This is the reverse process of distributing, not all quadratics can be factored.

$$x^2 + 8x + 7 = 0$$

$x^2 + 8x + 7 = 0$

$(x + 7)(x + 1) = 0$

$x + 7 = 0$ OR $x + 1 = 0$

$x = -7$ $\quad x = -1$

Solution Set $\{-7, -1\}$

Finding Zeros continued - You can also find zeros by using the quadratic formula.

The quadratic formula works on all quadratics.

$2x^2-3x-4$

$a=2 \quad b=-3 \quad c=-4$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$x=\frac{-(-3)\pm\sqrt{-3^2-4(2)(-4)}}{2(2)}=\frac{3\pm\sqrt{9+32}}{4}$$

$$x=\frac{3+\sqrt{41}}{4} \quad \text{or } x=\frac{3-\sqrt{41}}{4}$$

$x=2.35 \qquad \text{or } x=-0.851$

3b. Cubic Functions

Definition: Cubic functions are any functions of the form $y = ax^3 + bx^2 + cx + d$, where a, b, c, and d are constants, and a is not equal to zero. Cubics are polynomial **functions** with the highest exponent equal to 3.

Representations:

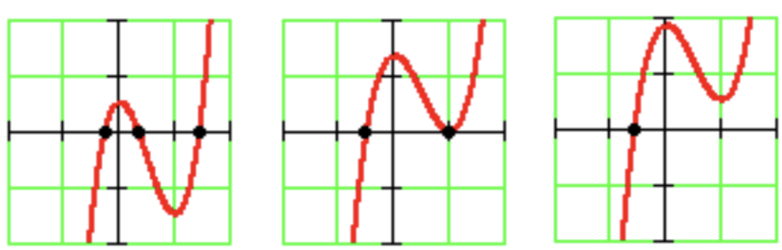
General Form:

$f(x) = ax^3 + bx^2 + cx + d$

where $a \neq 0$

$f(x) = 2x^3 - 5x^2 + 3x + 8$

Graphs: Cubic functions can have 3 roots, 2 roots or just 1 root.

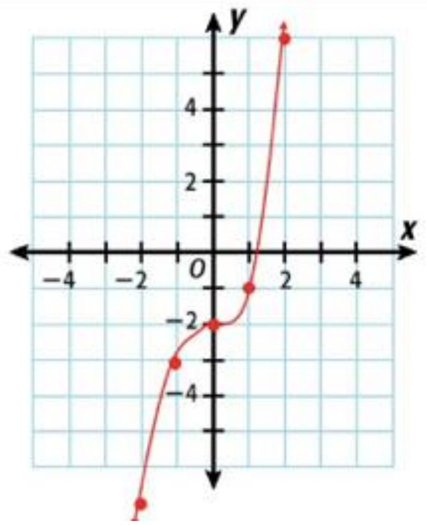


3 roots   2 roots   1 root

Table and graph of Cubic:

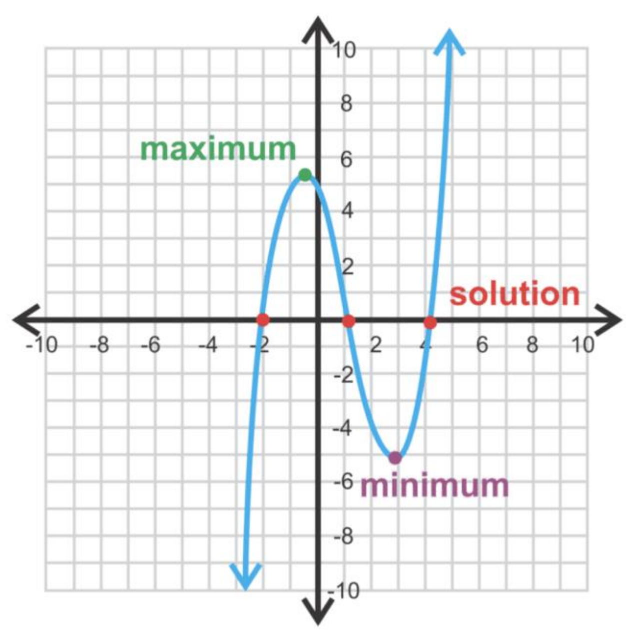
C. $y = x^3 - 2$

x	$x^3 - 2$	y
−2	$(-2)^3 - 2$	−10
−1	$(-1)^3 - 2$	−3
0	$(0)^3 - 2$	−2
1	$(1)^3 - 2$	−1
2	$(2)^3 - 2$	6



Cubic Maximum and Minimums -

Because of the shape of the graph of a cubic function, you often have the two turning points between the graph going up on one side and down on the other. These two internal turns create a relative maximum and a relative minimum.



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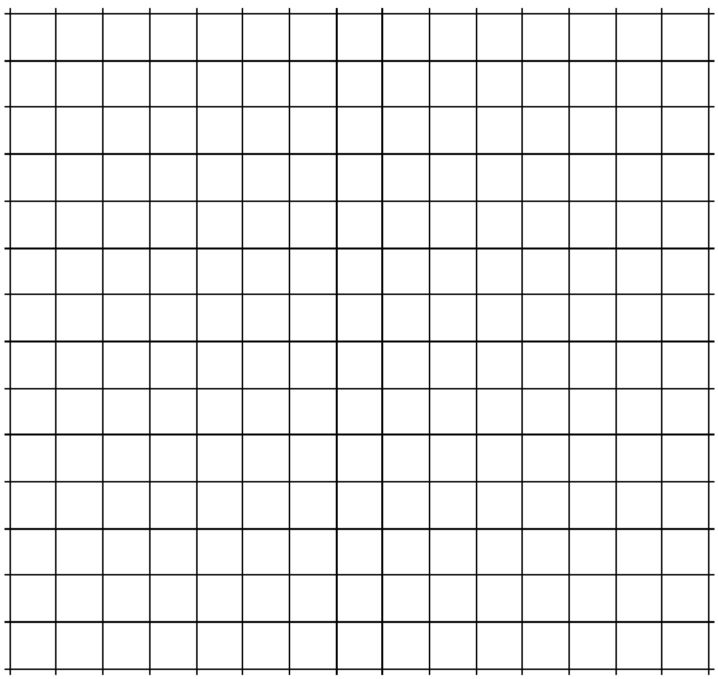
The Functions of Algebra 2

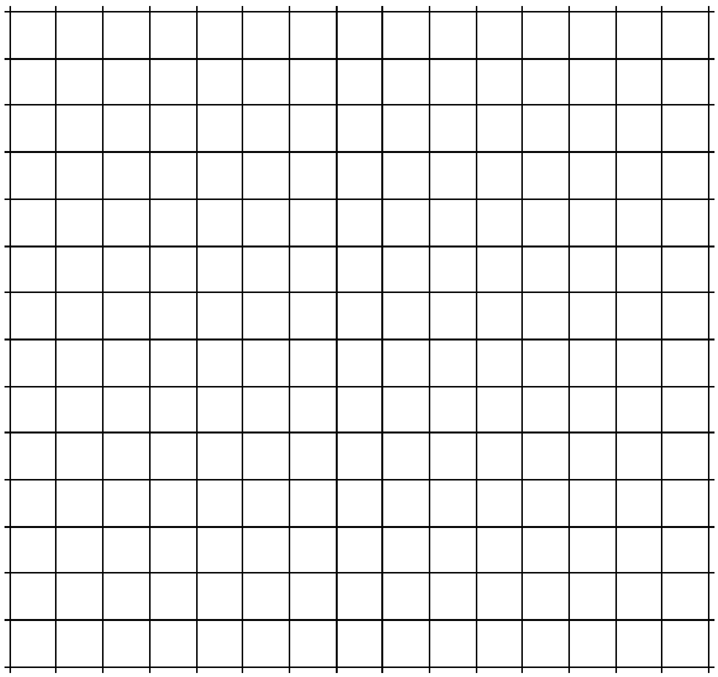
1. For each function, write a scenario and sketch a graph that might be modeled by the function.

   a. linear
   $y = 60x$



b. quadratic
$y = -16x^2+32x$

c. exponential
$y = 2^x$



2. Simplify each expression.

   a. $3\boldsymbol{x}^2 + 2\boldsymbol{x} - 3\boldsymbol{xy} - 5\boldsymbol{x} - \boldsymbol{x}^2 + 5\boldsymbol{xy} + 2\boldsymbol{y}^2$

   b. $\boldsymbol{x}(3 - 4\boldsymbol{x}) - 2(\boldsymbol{x}^2 + 5\boldsymbol{x}) + 6$

   c. $(\boldsymbol{x} - 3)(\boldsymbol{x} + 4)$

3. Consider the quadratic function $h(t) = -16t^2 + 150t + 5$. What is the degree of this polynomial function? What is the value of the leading coefficient and the constant term?

4. Can a linear function also be considered a polynomial function? Why or why not?

5. If you add polynomial expressions together, will the sum always be a polynomial expression? When you subtract polynomial expressions, will the difference always be a polynomial expression? Justify your answers.

6. Find the indicated products.

   a. $(\boldsymbol{x} + 2)(\boldsymbol{x} - 6)$

   b. $(\boldsymbol{x}^2 - 4)(\boldsymbol{x} - 6)$

   c. $(\boldsymbol{x} + 1)(\boldsymbol{x} - 1)(\boldsymbol{x} + 3)$

   d. $(\boldsymbol{x} + 2)(\boldsymbol{x} - 1)(\boldsymbol{x} + 3)$

7. A rectangular cooler is 18 inches long, 12 inches wide, and 12 inches deep.

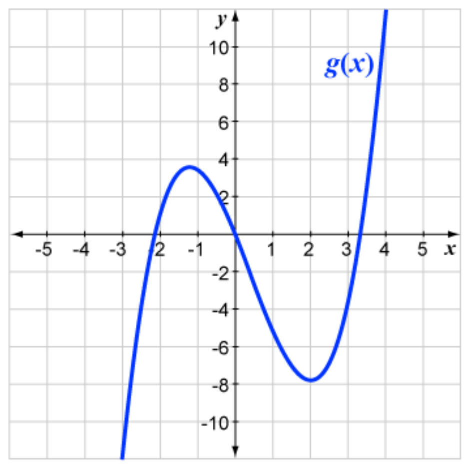
   a. What is the volume of the cooler?

   b. If each dimension is increased by $\boldsymbol{x}$ inches, what would be the new volume?

8. What type of expression do you get when you multiply three linear expressions?

9. Which of the cubic functions represented here has the greatest relative maximum? Explain how you know.

x	$f(x)$
-5	-147
-4	-72
-3	-25
-2	0
-1	9
0	8
1	3
2	0
3	5
4	24
5	63



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Zeros and Factors of Polynomials

Zeros, Roots and Factors:

Zeros, Roots, and Factors are related, but they are not the same thing.
Zeros are the actual points where the graph of a function crosses the x axis.
Factors, when multiplied together, give you your original product.
Roots are the x values of your function that make the function zero.

Factors, Roots, Zeros

For our *Polynomial Function*:

$$y = x^2 + 2x - 15$$

The *Factors* are: $(x + 5)$ & $(x - 3)$

The *Roots/Solutions* are: $x = -5$ and 3

The *Zeros* are at: $(-5, 0)$ and $(3, 0)$

Once you have the factors, you set each factor equal to zero to find the values of x, which are the roots. In the above example the factors are (x+5) and (x- 3) are the factors.

x + 5 = 0. Subtract 5 from both sides and you get x = -5
x – 3 = 0. Add three to both sides and you get x = 3.

-5 and 3 are your roots. When you plug the roots back into the x of your original equation you find that y = 0, and the zero's are the ordered pairs of your roots and zero, (-5, 0) and (3, 0).

Finding Factors of Quadratics by Factoring

An example of the general format of a quadratic is $x^2 + 5x + 4$. Using this quadratic, we will find the factors. The process of finding the factors is the process of finding two quantities that multiply together to create this quadratic.

We know that we need to find the factors of the last term, and that the middle coefficient is the sum of the factors of the last term. By making a chart, we can see how this this.

Finding Factors of Quadratics Continued

Using the quadratic $x^2 + 5x + 4$ as our example, we create a chart:

The factors of the last term are listed in the chart.

The sum of those factors are in the second column.

The sum of the factors in the first column match the Coefficient of the second term, so the factors are:

(x + 1) and (x + 4) So our roots are x = -1 and -4. When we check our factors by multiply these two factors we get the original quadratic.

$(x + 1) * (x + 4) = x^2 + 4x + 1x + 4 = x^2 + 5x + 4$

$x^2 + 5x + 4$

Factors of 4 (last term)	Sum of factors (middle coefficient)
1 x 4	1 + 4 = 5
-1 x -4	-1 + -4 = -5
2 x 2	2 + 2 = 4
-2 x -2	-2 + -2 = -4

Factoring works well when the number in front of the squared term is 1. It becomes more complicated when there is a number (coefficient) in front of the squared term. There are also times when the factors are not whole numbers, and it becomes difficult to use this method. A second process to find the factors of a quadratic is the quadratic equation, and the quadratic equation always works.

Finding Factors of Quadratics by Quadratic Formula

The quadratic formula is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Using our quadratic example from above, $x^2 + 5x + 4$. Our a = 1, b = 5, and c = 4. When we plug these numbers into our quadratic formula we get:

$$x = \frac{-5 \pm \sqrt{5^2 - 4(1*4)}}{2(1)}$$ Simplify the radical $$x = \frac{-5 \pm \sqrt{25 - 16}}{2}$$ or $$x = \frac{-5 \pm \sqrt{9}}{2}$$

$$x = \frac{-5 \pm 3}{2}$$ Simplified, x = –1 or – 4. These are our roots.

Finding Factors of Cubics by Grouping

Using the cubic function $x^3 - 6x^2 + 12 = 12x$ as our example, we use grouping as our process to find factors.

Solve $x^3 - 6x^2 + 12 = 2x$.

SOLUTION

$x^3 - 6x^2 - 2x + 12 = 0$ Rewrite in standard form.

$(x^3 - 6x^2) + (-2x + 12) = 0$ Group terms.

$x^2(x - 6) + (-2)(x - 6) = 0$ Factor each group.

$(x^2 - 2)(x - 6) = 0$ Use distributive property.

$x^2 - 2 = 0$ or $x - 6 = 0$ Use zero product property.

$\quad +2 \quad +2 \qquad\qquad +6 \quad +6$

$\sqrt{x^2} = \sqrt{2} \qquad\qquad x = 6$

$x = \sqrt{2}, x = -\sqrt{2}, x = 6$ Solve for x.

A second example of factoring a cubic using the function $2x^3 - 14x^2 = -24x$

Solve $2x^3 - 14x^2 = -24x$.

SOLUTION $2x^3 - 14x^2 + 24x = 0$ Rewrite in standard form.

$2x(x^2 - 7x + 12) = 0$ Factor common monomial.

$2x(x - 4)(x - 3) = 0$ Factor trinomial.

$2x = 0$ or $x - 4 = 0$ or $x - 3 = 0$ Use zero product property.

$x = 0, x = 4, x = 3$ Solve for x.

ANSWER

The solutions are 0, 4, and 3. Check these in the original equation.

Converting Quadratics from Standard form to Vertex form

You will recall that out two forms of our functions for quadratic equations are:

general form

$$f(x) = ax^2 + bx + c$$

vertex form

$$f(x) = a(x-h)^2 + k$$

To convert a quadratic equation from the general form to the vertex form, you do the following:

Standard Form: $y = ax^2 + bx + c$ Vertex Form: $y = a(x - h)^2 + k$

Convert from Standard Form to Vertex Form:

$y = ax^2 + bx = c \implies y = a(x - h)^2 + k$

know a, b, c want a, h, k

a = a

$x = \frac{-b}{2a}$ = h

Solve for y = k

Substitute the values and rewrite.

Example 1:

$y = 8x^2 - 16x + 27$ We know *a*, *b*, *c* and want *a*, *h*, *k*

$a = 8$ ⟸ ***a*** is the coefficient of the x^2 term

$h = x = \frac{-b}{2a} = \frac{-(-16)}{2(8)} = \frac{16}{16} = 1$ ⟸ use the formula to find the value of ***h***

$k = y = 8(1)^2 - 16(1) + 27 = 8 - 16 + 27 = 19$ ⟸ substitute the value found for ***h*** into the original equation and solve for ***k***

$y = 8(x - 1)^2 + 19$

Example 2:

$y = 5x^2 - 40x + 67$ We know *a*, *b*, *c* and want *a*, *h*, *k*

$a = 5$ ⟸ ***a*** is the coefficient of the x^2 term

$h = x = \frac{-b}{2a} = \frac{-(-40)}{2(5)} = \frac{40}{10} = 4$ ⟸ use the formula to find the value of ***h***

$k = y = 5(4)^2 - 40(4) + 67 = 80 - 160 + 67 = -13$ ⟸ substitute the value found for ***h*** into the original equation and solve for ***k***

$y = 5(x - 4)^2 - 13$

Converting Quadratics from Vertex form to Standard form

$$y = a(x - h)^2 + k \quad \Longrightarrow \quad y = ax^2 + bx = c$$

<u>Example 1:</u>

$y = 5(x + 2)^2 - 9$

$y = 5(x + 2)(x + 2) - 9$ ⟸ Rewrite $(x + 2)^2$

$y = 5(x^2 + 4x + 4) - 9$ ⟸ Simplify $(x + 2)(x + 2)$

$y = 5x^2 + 20x + 20 - 9$ ⟸ Distribute the 5

$y = 5x^2 + 20x + 11$ ⟸ Combine Like Terms

<u>Example 2:</u>

$y = -3(x - 4)^2 + 7$

$y = -3(x - 4)(x - 4) + 7$ ⟸ Rewrite $(x - 4)^2$

$y = -3(x^2 - 8x + 16) + 7$ ⟸ Simplify $(x - 4)(x - 4)$

$y = -3x^2 + 24x - 48 + 7$ ⟸ Distribute the -3

$y = -3x^2 + 24x - 41$ ⟸ Combine Like Terms

Zeros and Factors of Polynomials

1. Complete the following table to show possible dog kennels with a perimeter of 30 meters.

P(m)	*Length* (m)	*Width* (m)	*Area* (m^2)
30		1	
30		2	
30		3	
30		4	
30		5	
30		6	

2. If the width of the kennel is x meters, then what is the length in terms of x?

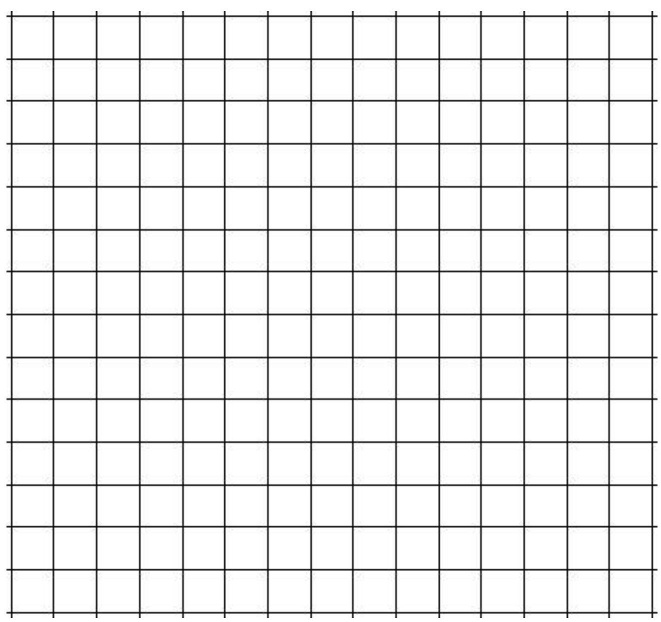
3. Write an expression for the area of the above rectangle in terms of x. Then write an equation you could solve to find the dimensions of a rectangle with an area of 50.

4. The height of a punted football is modeled by the following quadratic function:

$$\boldsymbol{h}(\boldsymbol{t}) = -5\boldsymbol{x}^2 + 20\boldsymbol{x} - 15.$$

a. Rewrite the function rule in vertex form.

b. Sketch a graph of the function.



5. Solve the following quadratic equation by factoring.

$\boldsymbol{x}(15 - \boldsymbol{x}) = 50$ (hint: you might need to distribute the x first)

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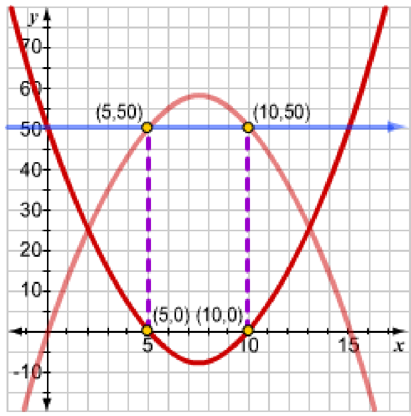
Zeros and Factors of Polynomials

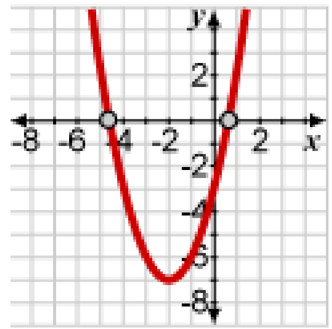
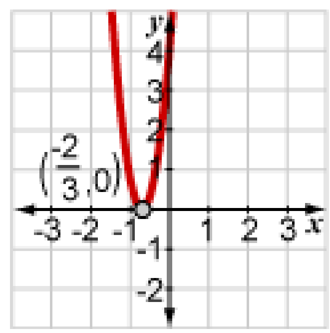
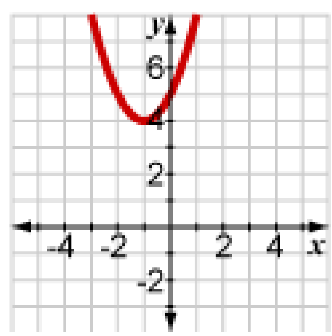
6. You have solved the quadratic equation $\boldsymbol{x}(15 - \boldsymbol{x}) = 50$ using two methods: graphing and factoring. How are these two methods related?



$$x(15 - x) = 50$$
$$x^2 - 15x + 50 = 0$$
$$(x - 5)(x - 10) = 0$$
$$x = 5 \text{ or } x = 10$$

7. Solve the equation $x^2 - 15x + 50 = 0$ using the quadratic formula.

8. Complete the table to represent the three possibilities for the solutions to quadratic equations.

Discriminant	Number and type of root(s)	Example sketch
$b^2 - 4ac > 0$	Two real roots	
$b^2 - 4ac = 0$		
$b^2 - 4ac < 0$		

9. Fill in the blanks to factor the polynomial $x^3 - 2x^2 + 3x - 6$ by grouping the terms using parentheses.

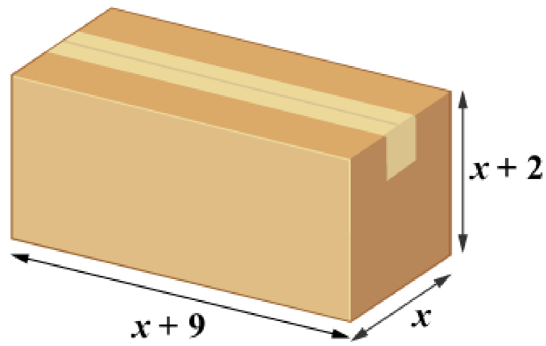
x^2	$(x^3 - 2x^2) + (3x - 6)$	$x + 3$	$x^2 - 2$
3	$x^2 + 3$	$x - 2$	x

Group the first two terms and the last two terms together using parentheses.	
Factor out the GCF from each group.	
Factor out the common binomial.	

10. Factor the following polynomials by grouping using the method of your choice.

a. $6x^3 - 4x^2 - 9x + 6$

11. An equation that could be used to determine the dimensions of this box, given that its volume is 40 cubic centimeters is $\boldsymbol{x}(\boldsymbol{x} + 2)(\boldsymbol{x} + 9) = 40$.



    a. What strategies could you use to solve the equation?

12. Rewrite each expression in factored form.

a. $3x^3 + 18x^2 + 9x - 30$

b. $4x^3 + 12x^2 - 25x - 75$

c. $x^3 + 6x^2 + 4x + 24$